



AI SUPPLY CHAIN INTELLIGENCE

Delay Detection & Solution BOT (SCI-BOT)



CUSTOMER INTERACTION



Customer opens SCI-BOT

Enter Shipment ID OR Customer Details



Mobile Number



Email Address



Order ID



Customer Name



Ask Shipment Question



LANGGRAPH WORKFLOW ENGINE



1
Input Node



2
Customer
Lookup
Node



3
RAG
Retriever
Node



4
Shipment
Tool
Node



5
Supplier
Tool
Node



6
Weather
Tool
Node



7
Risk
Analysis
Node



8
LLM
Node



9
Output
Node



LANGCHAIN ECOSYSTEM



Prompt
Templates



Tool
Calling



Output
Parser



Conversation
Memory



State
Management



LLM
Integration



RAG PIPELINE



Documents



Text
Splitter



Sentence
Transformers
Embeddings



FAISS
Vector
Database



Relevant
Knowledge
Retrieval



EXTERNAL TOOLS



Customer
Lookup
Tool



Shipment
Tool



Supplier
Tool



Weather
Tool



Risk
Assessment
Tool



GROQ LLM

GROQ
POWERED

AI analyses retrieved knowledge + live tool results
and generates the final business report.



FINAL AI RESPONSE



Customer
Information



Shipment
Information



Executive
Summary



Shipment
Status



Weather
Impact



Supplier
Analysis



Risk
Assessment



Recommended
Actions



Estimated
Business Impact



contact.sugumarai@gmail.com



AI SUPPLY CHAIN INTELLIGENCE

Delay Detection & Solution BOT



BOT NAME

SCI-BOT

Your AI Logistics Assistant



CUSTOMER INFORMATION



Customer ID : CUST1004
Customer Name : Divya
Phone Number : 9876543210
Email : divya@example.com
Order ID : ORD1004



SHIPMENT INFORMATION

Shipment ID : SHP1004
Status : **Delayed**
Supplier ID : SUP001
Warehouse ID : WH001
Destination City : Mumbai



SUPPLIER STATUS

Available

Supplier is active and inventory is sufficient.



WEATHER STATUS

Flood Warning

Severe weather conditions may impact delivery.



RISK LEVEL

High Risk

High risk of delay due to weather conditions.



ESTIMATED DELAY

48 Hours

Estimated delay in shipment delivery.

SHIPMENT STATUS REPORT



1 EXECUTIVE SUMMARY

The shipment status has been queried, and based on the available information, we provide an update on the current shipment status, weather impact, supplier analysis, risk assessment, recommended actions, and estimated business impact.



2 SHIPMENT STATUS

According to the live shipment information, the current shipment status is **Delayed**. As per the shipment policy, customers should receive updates whenever the shipment status changes, and they will be notified about this delay.



3 WEATHER IMPACT

The weather report indicates a **Flood Warning**, which, as per the delay rules, may delay shipments by **48 hours**. This weather condition is the likely cause of the current shipment delay.



4 SUPPLIER ANALYSIS

The supplier status is **Available**, indicating that the supplier is active and has not reported any inventory issues. However, it is essential to monitor the supplier's performance to ensure that they maintain at least **95% on-time delivery** as per the supplier guidelines.



5 RISK ASSESSMENT

The risk level is classified as **High Risk** due to the flood warning and the potential for further delays. It is crucial to closely monitor the situation and take proactive measures to mitigate any potential risks.



6 RECOMMENDED ACTIONS

- Notifying the customer about the delay and providing regular updates.
- Closely monitoring the weather conditions and supplier performance.
- Exploring alternative shipping routes or modes to reduce the delay.
- Reviewing the supplier's inventory levels to ensure they can meet the demand.



7 ESTIMATED BUSINESS IMPACT

The delay may result in a **48-hour delay** in shipment delivery, which could impact customer satisfaction and potentially lead to additional costs. However, by taking proactive measures and keeping the customer informed, we can minimize the business impact and maintain a high level of customer service.



AI-SUPPLY-CHAIN-INTELLIGENCE- DELAY-DETECTION-SOLUTION-BOT



AI POWERED MULTI-AGENT SUPPLY CHAIN INTELLIGENCE SYSTEM

Detect Delays | Analyze Risks | Recommend Solutions | Improve On-Time Delivery



ABOUT THE CHATBOT

AI-Supply-Chain-Intelligence-Delay-Detection-Solution-BOT is an intelligent AI assistant that provides real-time shipment tracking, delay detection, risk assessment, and actionable solutions using advanced AI, RAG, Multi-Agent system and external tools.

WHAT THIS CHATBOT DOES



Retrieves Shipment Information

Find shipment details using Shipment ID, Mobile Number, Email, Order ID or Customer Name.



Checks Weather Impact

Gets real-time weather information that may cause shipment delays.



Analyzes Supplier & Warehouse

Checks supplier availability, inventory status and warehouse congestion.



Detects Risks

Identifies possible risks like weather, supplier issues, warehouse congestion, traffic and mechanical failures.



Recommends Solutions

Provides intelligent recommendations to reduce delays and improve delivery performance.



Generates Professional Reports

Creates a detailed executive report with status, analysis, actions and business impact.

INPUT VALIDATION



All user inputs are validated before the AI system processes your request.

WHAT IS CHECKED?

VALIDATION	WHAT IS CHECKED?
Search Value	Ensures the search value field is not empty.
Question	Ensures the question field is not empty.
Shipment ID	Must start with "SHP" followed by 4 digits. Example: SHP1001
Mobile Number (Digits Only)	Allows only numeric characters (0-9).
Mobile Number (Length)	Must contain exactly 10 digits. Example: 9841629396
Email	Must be a valid email format. Example: user@gmail.com
Order ID	Must start with "ORD" followed by 4 digits. Example: ORD1001
Customer Name (Length)	Must contain at least 3 characters.
Customer Name (Characters)	Allows only letters (A-Z, a-z) and spaces.



If any validation fails, a clear error message is shown and the request is not processed further.

TECHNOLOGIES USED



LangChain



LangGraph



RAG



Memory



FastAPI



Groq



Python



Gradio



Secure • Reliable • Intelligent • Real-time



Email: contact.sugumarai@gmail.com